

Deep Fake Face Detection using Efficient Convolutional Neural Networks

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Abstract—Deep fake content, especially in images and videos, is spreading at an unprecedented rate. Fake content are generated with the help of advanced deep learning (DL) algorithms such as GANs, autoencoders, and variation autoencoders. This fake content spreads misinformation, causing a severe impact on society by degrading the trustworthiness of content on social media. Mitigating the risk of these techniques can be done by utilizing the power of one of the DL models, which is CNN. This research study focuses on analysing various Deep-Fake (DF) identification techniques that are trained on various datasets with a small number of samples. The proposed work demonstrates an efficient CNN model and three other CNN pre-trained models through transfer learning on a large dataset available on Kaggle, consisting of 140k images of faces. The proposed CNN model achieved a high accuracy of 96%, while DenseNet121 reached 97%. Both EfficientNetB0 and MobileNet demonstrated even higher performance, each achieving an accuracy of 98% within 15 epochs.

Keywords—Deep-Fakes (DF), Deep Learning (DL), CNN: Convolutional Neural Networks, GAN: Generative Adversarial Neural Network, Autoencoders.

I. INTRODUCTION

Nowadays, technology is available to everyone, and people are being exposed to the latest advancements. Recent times have witnessed a drastic technological advancement in artificial intelligence making machines to think and act according to the instructions given to them. The development of Deep Fakes (DFs) employ deep learning algorithms to produce realistic illustration but artificial media without the knowledge of the people portrayed, has been done by using AI based image and video editing tools. These algorithms can replicate human behavior and speech by utilizing open-source deep learning frameworks and the wealth of online material. This presents a huge challenge in the fight against the spread of fake content.

Deep Fake (DF) technology has both positive and negative aspects. On the positive side, it can generate morphed content for the entertainment industry, such as de-aging actors or bringing deceased ones back, as well as for voice acting, dubbing, and creating realistic training content for doctors. It can help teach computers to spot fake content, recreate historical events, and enhance old audio and video recordings. However, the misuse of DF technology poses

significant risks, such as spreading misinformation through fake videos of public figures, impacting elections and public trust, enabling phishing attacks, and creating non-consensual adult content. Detecting and stopping fake content remains challenging, necessitating better methods to distinguish real from fake and educating people on recognizing the differences. This work focuses on DFs, which use DL, a type of artificial intelligence that employs multi-layer artificial neural networks to imitate the functioning of the human brain. DL algorithms extract complex information from data by processing it gradually. To be able to recognize complex patterns, these models must be trained on large datasets. DL helps AI understand complex facial emotions, mouth movements, and voice patterns in source footage and high-resolution photos, which helps AI create convincing DFs. These algorithms mimic the target person's speech patterns, realistic emotions, and facial features with great accuracy.

Some DL techniques to generate DFs are described below.

Generative Adversarial Networks (GANs) generate artificial images or videos using a two-player game approach: the generator creates realistic synthetic data, while the discriminator evaluates its authenticity. The generator iteratively improves based on the discriminator's feedback until the synthetic data is indistinguishable from real data. In DeepFakes, this technique manipulates media to convincingly alter faces or voices.[13] [10] [8].

Variational Autoencoders (VAEs) and Autoencoders are both types of DL neural networks used for compressing and reconstructing data. VAEs focus on capturing the essence of the original data in a smaller space, while in DFs, VAEs can be employed to encode both the original and target faces, allowing for subtle changes. Autoencoders, on the other hand, learn from large data to make realistic faces or voices, even when there are limited samples, and are also used to enhance image compression by generating compact versions of images [12]. Both are crucial for creating effective DFs.

Additionally Facial Recognition Algorithms are used to identify and track people's faces in images or videos. This helps in selecting the target face for manipulation in DFs [7].

On the detection side, successfully distinguishing between real and fake media by extracting and analyzing critical elements like facial expressions, speech patterns, and spatial-temporal signals, DL is an essential tool in the fight against the spread of DF content. Large datasets, including both actual and fake information from multiple sources such as public repositories and social media platforms, are needed to efficiently train DL models. To reduce prediction errors and increase detection accuracy, these models are carefully trained via supervised learning.

A variety of DL techniques can be used in DF detection, including convolutional neural networks (CNNs), recurrent neural networks (RNNs), and generative adversarial networks (GANs). CNNs are efficient in image-based DF detection, while RNNs are effective for analyzing temporal sequences in videos [10] [15]. Additionally, GANs, traditionally used for DF creation, can also be used for detection by learning to discriminate between real and manipulated media.

The main objective of this study is to develop an automated system for detecting and classifying deep fake faces using CNN. Specifically, this study aims to:

1. To implement efficient CNN models to accurately distinguish real vs. fake images with high computational efficiency. Use a large dataset of 140,000, with 70k being manipulated images and 70k being real images, to improve model generalization and robustness, as more data helps the model learn better.
2. To achieve effective training with fewer epochs, employing efficient CNN models to minimize computational resources and accelerate the training process.

This research study was organized in six sections 1. Introduction, 2. Literature Review, 3. Proposed System, 4. Results and Analysis, 5. Conclusion, 6. References.

II. LITERATURE REVIEW

Detecting DF is not a new paradigm; several research studies have been conducted on DF detection in both images and videos. Many works have been published on this topic. Mostly, the data used to train these models were generated by GANs because these networks have the ability to automate the generation of DF images, which are difficult for human eyes to detect. However, there are several other techniques present to detect the images generated by GANs. One such technique is using CNN for classifying whether an image is real or fake, as CNNs are meant to classify images. Many authors have employed CNN architecture to automatically learn hierarchical representations of features from raw image pixels.

Several research studies have focused on detecting DF face images using CNN models as a means to detect fake images. One such paper is [1], the authors aim to utilize the effectiveness of CNN models to detect fake images. The aims to achieve higher accuracy in classifying fake images, mainly by reducing the cost and effort of detection. The dataset of 9000 images undergoes data augmentation, and

fed into their custom CNN model, which is a 13-layer architecture, alongside some pre-trained models are used. The ResNet50 model performed the best with 99.18% accuracy. Their custom CNN model achieved 95.21% accuracy. Other models like MobileNet and InceptionV3 also showed over 98% accuracy.

In [2], the authors used the “real-and-fake-face-detection-dataset” from Kaggle, pre-processed with data augmentation. They employed a custom CNN model with 4 Conv2D, 4 Maxpooling, and ReLU activation layers, achieving 98.46% training and 99.44% testing accuracy after 150 epochs. To address limitations in [2] regarding video detection, the authors proposed [3], where they used the same CNN model and tested on videos instead of images. They applied the Haar Cascade method to extract facial regions from video frames, and if any frame was predicted as fake, the entire video was classified as fake. Their method correctly identified 91 out of 100 videos from the DF Detection Challenge dataset.

The custom CNN model in [4] consists of 5 convolutional blocks. This paper uses Flickr’s 140k dataset which is present in Kaggle is used to train and validate the model. Data augmentation was applied on the dataset. The dataset consists of color images with 3 channels which were converted to grayscale with just 1 channel to simplify the training process, the proposed CNN model achieved an accuracy of 91.82%, F1 score of 91% and a precision is 81% in detecting deep fake faces.

The paper [5] the authors have utilized the potential of pre-trained models to detect fake images to that of real once, firstly they have applied Error Level Analysis to the images in their dataset which is a technique that is used to identify whether an image has been modified or not based on the JPEG compression levels. The ELA images were sent to those pre-trained models for training, the extracted features were then classified using SVM and KNN with hyperparameter optimization. The highest accuracy of 89.5% was achieved using the ResNet18 architecture and the KNN classifier with optimized hyper parameters. GoogLeNet achieved an accuracy of 81% with KNN, while SqueezeNet obtained 69.4% accuracy with SVM.

The custom CNN model in [6] is trained detect the DF videos. The authors have created a dataset from videos these images were then pre-processed by using data augmentation, then sent to their custom model for training, various optimization functions and loss functions were used for training, then the model was run on testing and validation data. To improve accuracy, additional parameter and hyperparameter tuning was done. Finally, the model was run on a sample video, and classes were identified in each frame of the video.

The authors of study [7] used a layered approach, first identifying the person's face in the picture using already-existing facial recognition networks. After that, CNN is used to extract face features, which are subsequently fed to LSTM model to examine facial modifications and temporal sequences between frames. The last stage is a Recycle-GAN, which combines spatial and temporal data using

GANs. The study utilizes the FaceForensics++ dataset. For this dataset, the results achieve a good detection accuracy of above 99%. The authors recognize the difficulty of pre-processing image data to highlight relevant features and the benefits of removing high-frequency components and unnecessary noise using methods like Gaussian Blur and Gaussian Noise.

XceptionNet which is a specialized CNN model designed to detect fake images [8], inspired by Inception modules and introduces by Google researchers. It utilizes a unique architecture with 36 convolutional layers grouped into 14 modules. They have used depth wise separable convolution layers. This change and the Inception design offer superior feature extraction with lower computational costs. They have used Flickr's dataset, these images are preprocessed by resizing and converting to the YCbCr color system, which provides better accuracy for detecting fake images. During feature extraction, a modified version called MTXception Net is used, incorporating SeparableConv2D and MaxPooling2D layers for enhanced accuracy and reduced parameters.

The technique for detecting fake faces was studied by the authors in the paper [9] uses the VGG19 CNN model, this model is enhanced with two attention mechanisms, one the Layer-Integrated Channel Attention and Scaled Spatial Attention. LICA focuses on understanding the importance of data across channels, while SSA helps decide where to focus within an image. These mechanisms are integrated into the VGG19 network after the last convolutional layer. This refined output helps the model better detect manipulated faces. The trained model is tested on the DFFD dataset. Faces are pre-processed using FaceNet's MTCNN.

In paper [10] provides an overview of deep fake face detection techniques, including methods like GANs for creating fake images. Image detection methods, such as those use deep CNNs and pre-processing techniques like Gaussian blur to improve detection accuracy. Detecting fake videos, involves analysing spatiotemporal characteristics. Challenges include improving approaches to recognize increasingly realistic deep fakes and determining optimal detection architectures. Commonly used datasets like DFFD, VGGFace2, and FFHQ provide diverse training examples. Evaluation metrics like accuracy and precision are vital for assessing detection model performance.

In their research [11], the authors used three advanced methods to address fake audio and video. Method 1: MesoNet, a specially created deepfake image and video detection system that was inspired by InceptionNet, uses Meso-4 and a modified Inception module to analyze image attributes and identify face manipulation. It gains knowledge from DF datasets by highlighting key areas of images for classification. Method 2: To identify discrepancies, SyncNet, a lip-sync system, integrates audio (MFCC values) and mouth images. To identify DFs, its ConvNet architecture calculates the separation between the audio and video outputs. Method-3: By extending EfficientNet, the Video Face Manipulation Detection using Ensemble of CNNs technique improves face manipulation

detection. It makes use of triplet Siamese training to increase accuracy and an attention mechanism to help it comprehend important aspects.

The paper [12] provides an overview of the current position of research in DFs, which are fake audio, video, or image content which were created through deep learning techniques to make it look as it is real. The study reviews machine learning (ML) and DL models like SVMs, CNNs, and LSTMs, with CNNs achieving up to 95.75% accuracy on datasets like Celeb-DF and DFDC. The paper also explores public datasets such as FaceForensics++, Celeb-DF, and DFDC used for training DF detection models, along with metrics like SSIM and FID to assess DF quality.

The study [14] provides an overview of the DF detection problem and the classification of DF detection methods into statistical models, machine learning (ML), and deep learning (DL). The authors have compared the performance of four models, namely MobileNetV2, InceptionV3, VGG16 and XceptionNet in DF detection. The authors trained these models on FaceForensics++ dataset which contains images related to DFs, Face2Face, FaceSwap, Neural Textures. The models were trained separately and evaluated based on training accuracy, average loss, testing accuracy, precision, recall, and F1-score.

The researchers in [15] have evaluated several state-of-the-art CNN models to detect DF videos. The study compared four CNN models, the authors have implemented a custom dataset from FaceForensics++. Each model was enhanced with the LSTM layer to capture sequential information. EfficientNetB7 has attained the top accuracy of 75%, then ConvNeXt Large with 67.17%, the ResNet-152 and MobileNetV3 have achieved slightly low accuracies of 61% and 61.33%, respectively. However, all models were overfitted, which indicate the need for improved generalization. The paper also shows the need to address issues like overfitting and improve model efficiency through better training data and advanced methods.

Despite significant advancements in deep fake detection, several research gaps persist in the existing literature. Most studies have either focused on custom CNN models or individual pre-trained models without a comprehensive comparison, limiting the understanding of their relative strengths and weaknesses. High accuracy in many models has often come at the cost of increased computational complexity, making them less practical for real-time applications. Additionally, some studies have not leveraged data augmentation effectively, resulting in models that lack robustness and generalization. The use of varied custom datasets and pre-processing methods has also impacted the reproducibility of results, creating challenges in standardizing evaluations. Overfitting remains a critical issue, with many models showing poor generalization across different datasets. Furthermore, there is a lack of direct comparison with multiple state-of-the-art methods in a unified framework, which is essential for a comprehensive evaluation.

III. PROPOSED METHODOLOGY

The paper focuses on developing a DF detection system to fight against the wide-spread of manipulated media content. The study uses several models and techniques to tackle the increasing worries about whether digital content is real and trustworthy. Beginning with data preprocessing, the proposed system applies data augmentation techniques to enhance diversity and robustness. The model definition stage plays a crucial role, with a focus on designing a DL architecture optimized for effective feature extraction and classification, the model is designed to detect minor irregularities that suggest the content has been artificially changed or modified.

The first step in the proposed system involves the acquisition of data from the “DF” dataset, which contains Real and Fake images. The dataset (‘deepfake-and-real-images’) under consideration consists of a total of 70k x 2 images for training, 19.6k x 2 images for Validation. The images are 256x256 JPG files and are intended to be used in the DF Detection Challenge on Kaggle. These images are structured into two directories, Fake and Real images. The dataset is designed for the purpose of DF detection.

The pre-processing pipeline shown in Figure 1 serves as a crucial preparatory stage in deep-fake detection, laying the groundwork for accurate and reliable model predictions. Firstly, the dataset is split into training, and validation sets, enabling efficient model training, performance monitoring, and unbiased evaluation of the model's generalization capabilities.

A. Data Pre-processing

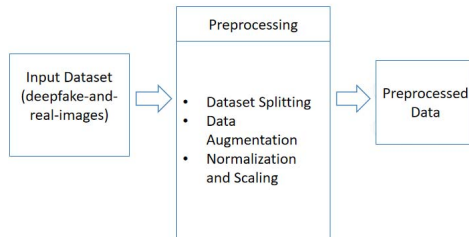


Fig 1: Pre-processing pipeline

Then Data augmentation techniques are employed on training set to enrich the dataset by generating diverse variations of the original images, enhancing the model's efficiency and generalization ability. After that, Normalization and standardization techniques are applied to ensure consistency in the input data distribution, mitigating numerical instabilities and expediting model convergence during training.

i. Strategic Dataset Splitting

The dataset is split into training and validation subsets. Training data teaches the model by revealing patterns, while validation data tests its performance on new data.

ii. Data Augmentation for Enhanced Diversity

The training set undergoes augmentation, diversifying it through techniques like rotations and flips as shown in Figure 2. This process enriches the dataset, helping the model in capturing the complexities of DF manipulation and preventing overfitting.

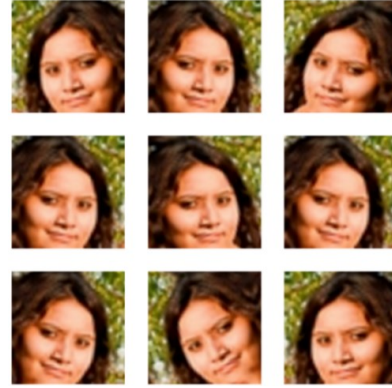


Fig 2: Data Augmentation

iii. Normalization and Scaling for Consistent Input

Prior to model training, images undergo pixel value scaling and normalization, ensuring even distribution and minimizing biases. This process reduces pixel values to the [0, 1] interval for numerical stability, leading to improved model performance and faster convergence during optimization.

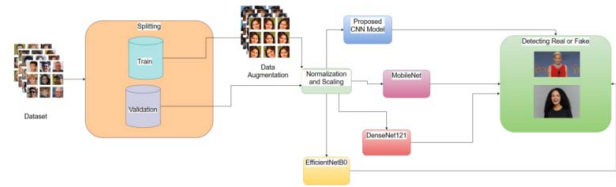


Fig 3: Proposed System Architecture

After preparing the data for training, the architecture shown in Figure 3 includes a proposed custom CNN model, as well as pre-trained models like MobileNet, DenseNet121, and EfficientNetB0, which are employed to learn and extract features from the images. The outputs from these models are used to detect whether a face is real or fake, facilitating robust deep-fake detection by combining the strengths of different CNN architectures.

B. Proposed CNN Model - 1

The model is a deep CNN designed for image classification tasks. It includes many advanced techniques such as depthwise separable convolutions, residual connections, batch normalization, and global average pooling. These components work together to efficiently learn and extract features from input images, the proposed CNN models architecture is shown in Figure 4.

i. Entry Block

The model starts with an entry block, which serves as the initial processing stage for the input image. In this block the input image, represented as "(180, 180, 3)" for RGB channels, is convolved with 128 filters of size 3x3 with a stride of 2. This initial convolution reduces the spatial dimensions of the image. The output of the Conv2D layer is then normalized using batch normalization. This step helps in stabilizing and accelerating the training process by normalizing the activations. Then a Rectified Linear Unit (ReLU) activation function is applied. ReLU introduces non-linearity to the network, allowing it to learn complex patterns in the data.

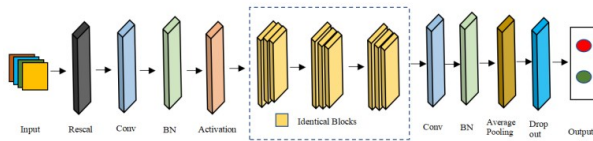


Fig 4: Proposed CNN Model Architecture

ii. Multiple Blocks

The model consists of multiple blocks, used for feature extraction and representation. These blocks are repeated for sizes '[256, 512, 728]', allowing it to learn features at different scales and complexities. These values represent the number of filters used in each depthwise separable convolutional layer within each block. Different sizes allow the model to learn features at different scales and complexities. This technique is often used in deep neural networks architecture to improve the model's ability to capture and represent various levels of information in the input data.

Components of Each Block

1. Activation (ReLU)

After each block's input, another ReLU (Rectified Linear Unit) activation is applied. It is a non-linear activation function that introduces non-linearity to the network. It helps in learning complex patterns and features in the data, allowing the network to model more complicated relationships.

2. Depthwise Separable Convolution

Two depthwise separable convolutional layers are stacked together in each block. These layers are designed to be computationally efficient, reducing the number of parameters and allow the model to learn spatial and cross-channel features effectively, it consists of the two layers 2.1 and 2.2.

2.1. Depthwise Convolution

The first part of the layer is a depthwise convolution, where each input channel is convolved separately with its own kernel. This operation applies a 3x3 convolution to each channel of the input independently. The goal is to learn spatial features within each channel.

2.2. Pointwise Convolution

The second part of the layer is a pointwise convolution, which combines the output of the depthwise

convolution. It uses 1x1 convolutions to mix information from different channels. This step helps in increasing the model's capacity to learn complex patterns and relationships across channels.

1. Batch Normalization (BN)

Batch normalization is applied after every depthwise separable convolutional layer. It is used to normalize the output of the previous layer across the batch dimension. It helps in stabilizing and accelerating the training process. By normalizing the activations, it makes sure that the model can train faster and do not overfit. The first BN layer occurs after the first depthwise separable convolution and the second one after the second depthwise separable convolution.

2. MaxPooling2D

After the two depthwise separable convolutions and BN, MaxPooling2D layer is applied. It downsamples the feature maps by taking the maximum value from a 3x3 window and moving with a stride of 2. It helps in finding the most important features and reducing the spatial resolution.

Each block in the model contains two depthwise separable convolutional layers then batch normalization and ReLU activation, all of these are important for feature extraction.

Residual Connections

In each block, a residual connection is introduced to help in training the network more effectively. It adds the output of the current block to a Conv2D layer's output from the previous block. This helps in reusing the important information again without being vanished, so helps stopping the vanishing gradient problem. The residual connection is created after the MaxPooling2D layer.

Within each iteration, the "current block" consists of two Depthwise Separable Convolutional layers, BN after each layer, ReLU activation after each BN, and finally the MaxPooling2D layer. The output from this "current block" is then combined with the residual connection from the Conv2D layer of the previous block to form the output for that iteration. This structure helps in learning residual functions, which can be beneficial for training deep neural networks.

iii. Final Layers

Towards the end of the model, after repeated blocks, the following layers are applied. A final depthwise separable convolutional layer with 1024 filters and a 3x3 kernel is applied. This layer further refines the learned features. BN is applied again, followed by ReLU activation. Then Global Average Pooling layer averages the feature maps across spatial dimensions, reducing each feature map to a single value. It helps in capturing the most important information from the feature maps. To prevent overfitting, a dropout layer with a dropout rate of 0.25 is included. Dropout helps in generalization. The final layer is a dense layer that produces the model's output. The model is used for binary classification; it has one unit with no activation. In our case,

it is binary classification because the model needs to distinguish between real and fake images.

C. Using Pre-Trained models for Binary Classification (DF Detection)

Pre-trained models were designed for multi-class classification, these models can also be adapted for binary classification tasks

This study utilizes pre-trained models which have been trained on a massive dataset like ImageNet. These model have already learned valuable feature representations.

i. Mobilenet (CNN Model - 2)

MobileNet, a lightweight CNN architecture designed by Google, is ideal for embedded and mobile vision applications due to its efficient use of depthwise separable convolutions, which significantly reduce computational complexity and the number of parameters while maintaining accuracy. This makes MobileNet perfect for resource-constrained contexts and a popular choice for real-time applications such as DF detection. Its scalable architecture effectively extracts features at various scales, making it versatile for tasks like object identification, semantic segmentation, and image classification.

ii. EfficientNetB0 (CNN Model - 3)

EfficientNetB0, developed by Google researchers, balances complexity and efficiency for superior performance by dynamically adjusting depth, width, and resolution through compound scaling. Integrating squeeze-and-excitation modules and convolutional layers with skip connections, it maximizes representational capacity and facilitates feature reuse. With low computational demands, EfficientNetB0 achieves high accuracy, making it ideal for image classification and other computer vision tasks in resource-constrained environments. Its scalability and efficiency make it a popular choice for real-time applications such as DF detection.

iii. DenseNet121 (CNN Model - 4)

DenseNet121, developed by Facebook AI Research (FAIR), features a dense connectivity structure where each layer passes data to all subsequent layers within its block, promoting feature reuse and mitigating the vanishing gradient problem. Comprising dense blocks and transition layers, it manages parameter growth and maintains spatial dimensions of feature maps, achieving cutting-edge performance in image classification without excessive complexity. DenseNet121's efficient use of parameters and ability to learn deep features with fewer parameters make it computationally efficient and versatile for various applications, including impressive performance in DF detection.

This paper has used three pre-trained models for DF detection, unfreezed all the layers of these pre-trained models while training which helps the network, unfreezing all the layers of a pre-trained CNN model is useful because

of the large dataset, such as 140k images, it allows the model to learn from the specific and important features and patterns in the dataset, added a Global Average Pooling layer, this layer compresses the information within a feature map by averaging its values across spatial dimensions (width and height), and a Dense layer for the classification of images in the dataset.

IV. RESULTS AND ANALYSIS

This research study presents a comparative performance analysis of four CNN models shown in Table 1. The experimentation results show better performance in terms of accuracy metric of different CNN models assessed on a classification job. Out of all the models evaluated, MobileNet showed a comparatively low loss of 0.0393 and an accuracy of 98.45%. With a little more loss of 0.0437, but an accuracy of 98.50%, EfficientNetB0 trailed closely behind. DenseNet121 had a loss of 0.0725 and a slightly lower accuracy of 97.18%. Finally, the accuracy of the proposed CNN model was 96.67% with a 0.0681 loss, despite its somewhat lower accuracy when compared to other models. The proposed model has been implemented using some advanced techniques unlike traditional CNNs such as depthwise separable convolutional layer and residual connections helps the model to better learn how to distinguish real images from that of fake images.

TABLE 1: EVALUATION METRICS

Models	Accuracy	Loss	Precision	F1-Score	Recall
MobileNet	0.9845	0.0393	0.9832	0.9865	0.9898
EfficientNetB0	0.9850	0.0437	0.9822	0.9822	0.9725
DenseNet121	0.9718	0.0725	0.9815	0.9715	0.9616
Proposed CNN Model	0.9667	0.0681	0.9735	0.9723	0.9613

Table 2 contains accuracy graphs depict the training and validation accuracy over 15 epochs for each model. The Proposed CNN Model exhibits a steady increase in both training and validation accuracy, stabilizing around an accuracy of 0.98. MobileNet shows a rapid rise in accuracy during the initial epochs, reaching around 0.98 for validation accuracy, with some fluctuations observed throughout the epochs. EfficientNetB0 shows good performance with training and validation accuracies close to 0.99, indicating a strong learning capability and generalization. DenseNet121, while demonstrating high initial training accuracy, experiences notable fluctuations in validation accuracy, particularly a significant drop around epoch 6, which indicates some instability during training.

The loss graphs provide insight into the convergence behavior of the models by plotting training and validation loss over the same epochs. The Proposed CNN Model and MobileNet both show a consistent decrease in loss, with validation loss stabilizing at a low level, suggesting effective learning. EfficientNetB0 maintains the lowest loss

values throughout the epochs, with both training and validation loss exhibiting minimal fluctuations, signifying robust performance. In contrast, DenseNet121 shows erratic behavior with substantial spikes in validation loss, particularly around epoch 6, which correlates with the observed drop in validation accuracy, indicating potential overfitting or training instability issues.

Overall, EfficientNetB0 outperforms the other models in terms of both accuracy and stability, while the Proposed CNN Model and MobileNet also demonstrate strong

performance with slight fluctuations. DenseNet121, despite high initial accuracy, requires further investigation to address its instability and improve generalization.

True color images (RGB) of size [180,180,3] is given as input for the proposed model, it predicts the image as real or fake.

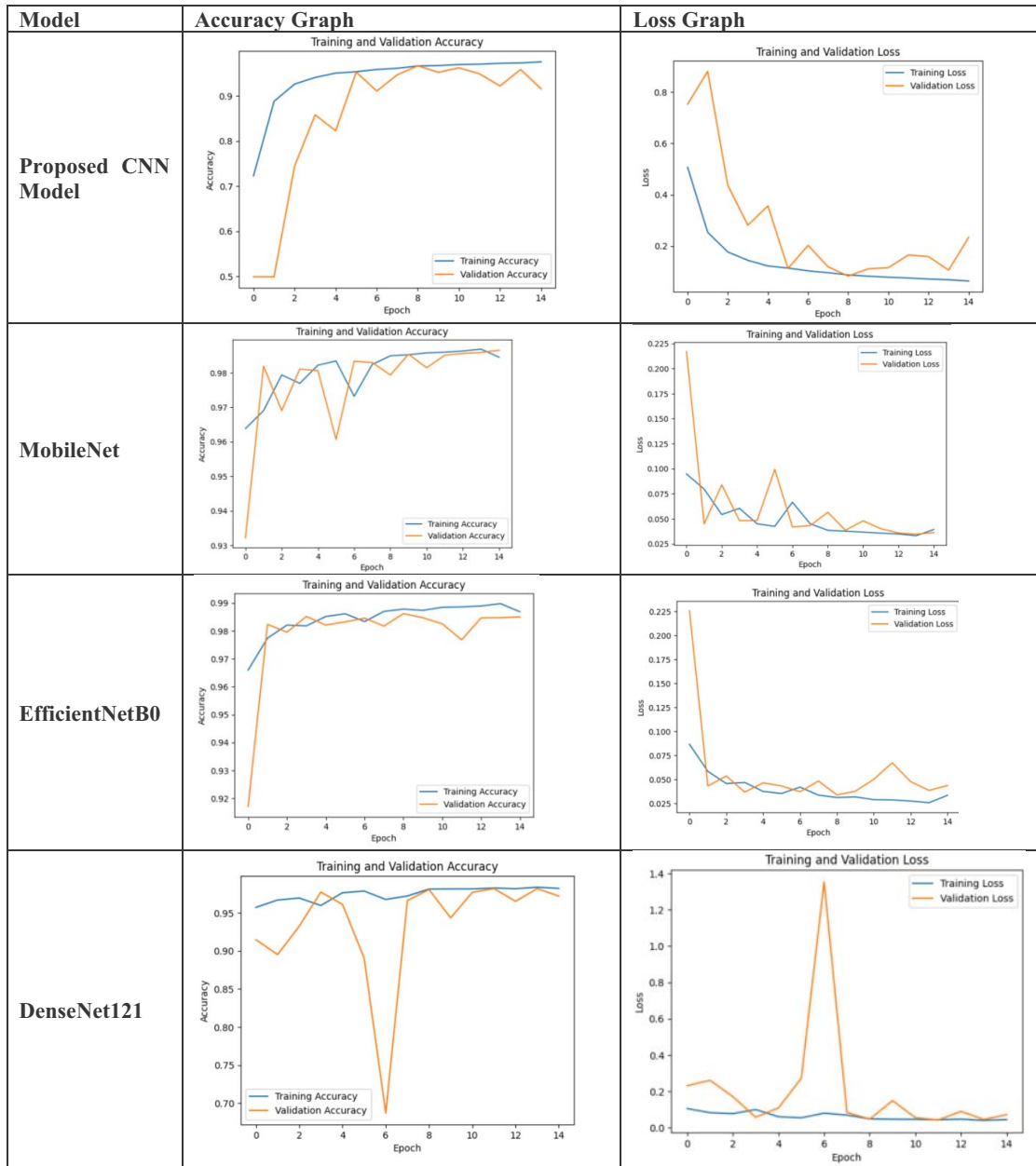


Fig 8: Predicting Fake image as Fake



Fig 9: Predicting Real image as Real

TABLE 2: ACCURACY AND LOSS GRAPHS



V. CONCLUSION

The Deep Fake detection system implemented in this paper has shown better performance in distinguishing real images from fake ones. This research study has considered four CNN models of which one is the proposed CNN model and rest are pre-trained models, trained on dataset containing images. This research study has utilized the power of pre-trained models by unfreezing all of its layers, which helps the proposed model to learn specific features in the considered dataset. The proposed model has achieved a good accuracy when trained on large dataset in less number of epochs.

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